

PRATEESH PATLOLLA

Data Scientist / ML Engineer

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SUMMARY

Applied Data Scientist with 5 years of experience delivering predictive forecasting, optimization, and AI-driven automation at enterprise scale. Built resilient, cloud-native ML data systems (Databricks, AWS, Spark) and designed conversational, tool-calling Ops Agents (LangGraph) that streamline capacity allocation and budget management. Skilled in A/B testing, causal inference, and cross-functional scenario modeling, with a track record of driving measurable ROI and operational efficiency across enterprise environments.

PROFESSIONAL EXPERIENCE

Griddynamics (Apple)
Senior Data Scientist

Austin, TX
Dec 2025 – Present

- Deployed a LangGraph tool-calling Ops Agent to automate budget and capacity inquiries for 40+ leaders, integrating 18 deterministic tools with memory to deliver sub-3-second live inference.
- Established strict LLMOps guardrails, gating agent releases behind 200+ automated golden-set evaluations to prevent hallucinations. Engineered a 1-click telemetry pipeline capturing full tool-call provenance for regression testing.
- Engineered a workforce forecasting pipeline projecting human-in-the-loop evaluation demand, optimizing vendor capacity allocation and enabling ad-hoc scenario modeling for budget trade-offs.
- Architected four scheduled ETL pipelines on Apple Cloud (ACAI) to consolidate daily spend workflows, establishing ACAI Conductor as the single source of truth for downstream Tableau dashboards and AI agents.
- Standardized volume-weighted forecast error metrics, resolving a 5x data discrepancy across legacy systems. Deployed automated telemetry to proactively flag stale intake forms and alert owners of budget overruns.

Toyota North America
Data Scientist

Dallas, TX
Feb 2023 – Nov 2025

- Led the design and POC of AskToyota, a GenAI chatbot using LangChain, FAISS, LangGraph, and LLaMA 2, deployed in secure VPC environments with IAM-based access control and TLS. Reduced stakeholder turnaround time by 30% by surfacing forecasts and KPIs on demand.
- Spearheaded Accessory Recommendation engine generating \$150M+ annual revenue using XGBoost, dynamic segmentation, and incentive-aware retraining, integrated A/B testing, and causal inference for robust evaluation.
- Built Toyota's Annual Production Optimizer (APO) using Gurobi to solve multi-objective LP problems (profit, volume, GHG score) across 14 global factories, optimizing \$700M+ in planning annually.
- Developed Smart Forecasting engine with PySpark and SARIMAX to predict dealer-level sales; enabled 3+ hours/month time savings per dealer and built self-serve analytics dashboards via Tableau.
- Productionized Pyspark and ML pipeline in Azure Databricks, integrating model registry, drift detection with auto-retraining, Airflow orchestration, and CloudWatch/Datadog monitoring for real-time alerts.
- Deployed real-time ETA API for dealer websites using ElastiCache Redis with SHA-1 for online inference; batch-trained models using PySpark and Parquet on S3. Packaged with Docker and deployed via Kubernetes.
- Blended causal inference and A/B testing frameworks with GenAI tools (Azure OpenAI, LangChain agents, Guardrails AI, Azure Cognitive Search, FAISS) on Azure Databricks to auto-explain accessory model KPIs, increasing dealer conversions by 20% and boosting stakeholder trust..

Amazon, Alexa AI
Data Scientist Intern

Santa Barbara, CA
May 2022 – August 2022

- Retrained and deployed Alexa's question categorizer using advanced NLP (multi-class classification, intent detection), improving query routing across 100+ categories via SageMaker.
- Used resampling and confidence intervals to evaluate and improve model confidence on rare query intents, increasing trust and accuracy for low-frequency categories in Alexa.
- Improved the generated success rate model's alignment with human-labeled success rate, reducing gap from ~20% to under 14%, enabling reduced annotation spend.
- Built reporting pipelines that pushed KPIs into Redshift and Quicksight to visualize key health metrics and rolling 3-month forecast trends to support weekly reviews with product stakeholders.

Cyient
Data Scientist

Hyderabad, India
Jan 2020 – December 2020

- Extracted road sign boards through object detection from terrestrial imagery to minimize manual effort in data annotation for North American-based clients.
- Achieved a hit rate of 92%, resulting in a saving of 12 FTEs.
- Designed and optimized AWS Glue ETL jobs to streamline data ingestion into a data warehouse, leading to significant improvements in data quality and processing speed.

EDUCATION

INDIANA UNIVERSITY BLOOMINGTON
Master's of Science in Data Science

Bloomington, Indiana, USA

GITAM UNIVERSITY
Bachelor's in Computer Science

Hyderabad, India

PROJECTS

Prateesh's Gen AI Resume Assitant

Chatbot Link: prateeshreddy.github.io

- Developed a scalable RAG pipeline using LangChain, FAISS, and OpenAI's embeddings, enabling context-aware, accurate Q&A about my professional background. Engineered advanced prompt templates and conversational memory, delivering enterprise-grade efficiency with 98% retrieval accuracy, 120 ms latency (p99), and inference costs below \$0.0005 per query.

Cognitive Search Engine by NLP

- Provided Cognitive search capability for **Eli Lilly and Company** to search against databases like FDA and EMA via natural language questions and return relevant results to help with accelerating regulatory submissions for Eli Lilly as an Intern in the Summer of 2021 using Hugging Face transformers models.

Research Scientist at Institutional Analytics, Indiana University

- Been part of the **guest lecture** for ILSZ 637 - Information Visualization by Noriko Hara in Spring 2022, a graduate-level course talking about data storytelling and data visualization. Built data pipelines with Tableau reports to analyze student performance, retention, and graduation trends, influencing key academic decisions.